A Scientific Autobiography

My original decision to devote myself to science was a direct result of the discovery which has never ceased to fill me with enthusiasm since my early youth—the comprehension of the far from obvious fact that the laws of human reasoning coincide with the laws governing the sequences of the impressions we receive from the world about us; that, therefore, pure reasoning can enable man to gain an insight into the mechanism of the latter. In this connection, it is of paramount importance that the outside world is something independent from man, something absolute, and the quest for the laws which apply to this absolute appeared to me as the most sublime scientific pursuit in life.

These views were bolstered and furthered by the excellent instruction which I received, through many years, in the *Maximilian-Gymnasium* in Munich from my mathematics teacher, Hermann Müller, a middle-aged man with a keen mind and a great sense of humor, a past master at the art of making his pupils visualize and understand the meaning of the laws of physics.

My mind absorbed avidly, like a revelation, the first law I knew to possess absolute, universal validity, independently from all human agency: The principle of the conservation of energy. I shall never forget the graphic story Müller told us, at his *raconteur's* best, of the bricklayer lifting with great effort a heavy block of stone to the roof of a house. The work he thus performs does not get lost; it remains stored up, perhaps for many years, undiminished and latent in the block of stone, until one day the block is perhaps loosened and drops on the head of some passerby.

After my graduation from the *Maximilian-Gymnasium*, I attended the University, first in Munich for three years, then in Berlin for another year. I studied experimental physics and mathematics; there were no professorships or classes in theoretical physics as yet. In Munich, I attended the classes of the physicist Ph. von Jolly, and of the mathematicians Ludwig Seidel and Gustav Bauer. I learned a great deal from these three professors, and I still retain them in reverent memory. But I did not realize until I came to Berlin that in matters concerned with science they had really just a local significance, and it was in Berlin that my scientific horizon widened considerably under the guidance of Hermann von Helmholtz and Gustav Kirchhoff, whose pupils had every opportunity to follow their pioneering activities, known and watched all over the world. I must confess that the lectures of these men netted me no perceptible gain. It was obvious that